

Perturbative Calculation of the Path Integral for the Anharmonic Oscillator

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1 Preliminary

The present goal is to provide a correction to the energy of a harmonic oscillator with an anharmonic perturbation, using exclusively the path integral approach. As a starting point we use the path integral form of the partition function from Swanson (4.14) (using a different convention for the integration limits for... reasons):

$$Z_\beta = \int dq_a \int_{q_a}^{q_a} \overline{\mathcal{D}}q \exp \left\{ - \int_{-\beta/2}^{\beta/2} d\tau \left[\frac{1}{2} m \dot{q}^2 + V(q) \right] \right\} \quad (1)$$

$$= \int dq_a \int_{q_a}^{q_a} \overline{\mathcal{D}}q \exp \left\{ - \int_{-\beta/2}^{\beta/2} d\tau \left[\frac{1}{2} m \dot{q}^2 + \frac{1}{2} m \Omega^2 q^2 + g q^4 \right] \right\} \quad (2)$$

where g is assumed to be small. To treat the quartic term perturbatively we use a Taylor series expansion like so (also taking the zero temperature limit so $\beta \rightarrow \infty$):

$$\begin{aligned} Z_\beta &= \int dq_a \int_{q_a}^{q_a} \overline{\mathcal{D}}q \exp \left[- \int_{-\infty}^{\infty} d\tau g q^4 \right] \exp \left\{ - \int_{-\infty}^{\infty} d\tau \left[\frac{1}{2} m \dot{q}^2 + \frac{1}{2} m \Omega^2 q^2 \right] \right\} \\ &= \int dq_a \int_{q_a}^{q_a} \overline{\mathcal{D}}q \left[1 - \int_{-\infty}^{\infty} d\tau g q^4 + \dots \right] \exp \left\{ - \int_{-\infty}^{\infty} d\tau \left[\frac{1}{2} m \dot{q}^2 + \frac{1}{2} m \Omega^2 q^2 \right] \right\} \\ &= Z_0 - \int dq_a \int_{q_a}^{q_a} \overline{\mathcal{D}}q \left[\int_{-\infty}^{\infty} d\tau g q^4 \right] \exp \left\{ - \int_{-\infty}^{\infty} d\tau \left[\frac{1}{2} m \dot{q}^2 + \frac{1}{2} m \Omega^2 q^2 \right] \right\} + \dots \\ &= Z_0 - \int_{-\infty}^{\infty} d\tau' g \int dq_a \int_{q_a}^{q_a} \overline{\mathcal{D}}q [q(\tau')]^4 \exp \left\{ - \int_{-\infty}^{\infty} d\tau \left[\frac{1}{2} m \dot{q}^2 + \frac{1}{2} m \Omega^2 q^2 \right] \right\} + \dots \\ &\equiv Z_0 \left(1 - \int_{-\infty}^{\infty} d\tau' g \langle q(\tau')^4 \rangle + \dots \right) \end{aligned} \quad (3)$$

In the first couple lines again we've just used a Taylor expansion, and then we've defined the unperturbed partition function Z_0 as the evaluation of (1) if $V(q) = m\omega^2 q^2/2$ (i.e. just without the perturbation). Then we've moved the time integral outside the path

integral, changed variable names for clarity (i.e. $\tau \rightarrow \tau'$) and then used the expectation value w.r.t. the unperturbed system, defined:

$$\langle f(q(t)) \rangle = \frac{1}{Z_0} \int dq_a \int_{q_a}^{q_a} \overline{\mathcal{D}}q [f(q(t))] \exp \left\{ - \int_{-\infty}^{\infty} d\tau \left[\frac{1}{2} m \dot{q}^2 + \frac{1}{2} m \Omega^2 q^2 \right] \right\} \quad (4)$$

We now direct our attention to the unperturbed term in (2), in particular taking a Fourier transform on q (this will make things simpler as we'll note later). This needs an integral from $-\infty \rightarrow \infty$ which is why we chose the integration limits like in (1)–(2) in conjunction with the zero temperature limit. This allows us to write:

$$\begin{aligned} \int_{-\infty}^{\infty} q(\tau)^2 d\tau &= \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega\tau} q(\omega) d\omega \right)^2 d\tau \\ &= \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-i(\omega+\omega')\tau} q(\omega) q(\omega') d\omega d\omega' d\tau \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \delta(\omega + \omega') q(\omega) q(\omega') d\omega d\omega' \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} q(\omega) q(-\omega) d\omega \end{aligned} \quad (5)$$

In the first line we've just used the definition of the Fourier transform, using $q(\omega)$ as shorthand for the Fourier transform of $q(\tau)$. The second line is straightforward, and in going to the third line we've performed the time integral to yield the δ -function, finally using it to collapse one of the integrals. It's worth noting that $q(-\omega) = q^*(\omega)$ (since $q(\tau)$ is real).

We can perform a similar set of steps to find (using the Fourier transform of a derivative):

$$\begin{aligned} \int_{-\infty}^{\infty} \dot{q}(\tau)^2 d\tau &= \int_{-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega\tau} (-i\omega) q(\omega) d\omega \right)^2 d\tau \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \omega^2 q(\omega) q(-\omega) d\omega \end{aligned} \quad (6)$$

We can thus write the action like so:

$$\begin{aligned} S &= \int_{-\infty}^{\infty} d\tau \left[\frac{1}{2} m \dot{q}^2 + \frac{1}{2} m \Omega^2 q^2 \right] \\ &= \frac{m}{4\pi} \int_{-\infty}^{\infty} (\omega^2 + \Omega^2) q(\omega) q^*(\omega) d\omega \end{aligned} \quad (7)$$

If we were to use this to find Z_0 , we would need to first change variables in the path integral from the time slices to frequency slices, but this is no big deal since there's no other q dependence outside of the measure, which is invariant because the Fourier transform is unitary.

This is even in ω and hence we can convert it to an integral from $0 \rightarrow \infty$ and add a factor of two, which gives the form:

$$Z_0 = \int dq_a \int_{q_a}^{q_a} \overline{\mathcal{D}}q \exp \left\{ -\frac{m}{2\pi} \int_0^\infty (\omega^2 + \Omega^2) q(\omega) q^*(\omega) d\omega \right\} \quad (8)$$

The reason for changing the bounds is to justify the measure, which integrates every point within $q(\omega)$ over the parameter space. This is a problem because $q(\omega)$ and $q(-\omega)$ are not independent variables, and the path integral cannot independently vary them, though with (8) this is no longer a problem.

We know from Wick's theorem that $\langle q^4 \rangle = 3 \langle q^2 \rangle^2$, and so our primary interest is finding $\langle q(\tau)^2 \rangle$. To do this note that $\langle q^*(\omega') q(\omega) \rangle = A^{-1}(\omega, \omega')$, where A is the quadratic coefficient matrix in the exponent, in this case:

$$A(\omega, \omega') = \frac{m}{2\pi} (\omega^2 + \Omega^2) \delta(\omega - \omega') \quad (9)$$

which is just the continuum analogue of a diagonal matrix (this is why we use frequency space), and the inverse is simply the reciprocal of the prefactor with the same δ -function. So, if we now consider the Fourier transform of $q(\tau)$, we can write:

$$\begin{aligned} \langle q(\tau)^2 \rangle &= \left\langle \left\{ \frac{1}{2\pi} \int e^{-i\omega\tau} q(\omega) d\omega \right\}^2 \right\rangle \\ &= \left\langle \frac{1}{4\pi^2} \int \int e^{-i\tau(\omega+\omega')} q(\omega) q(\omega') d\omega d\omega' \right\rangle \\ &= \frac{1}{4\pi^2} \int \int e^{-i\tau(\omega+\omega')} \langle q^*(-\omega) q(\omega') \rangle d\omega d\omega' \\ &= \frac{1}{4\pi^2} \int \int e^{-i\tau(\omega+\omega')} \frac{2\pi}{m(\omega^2 + \Omega^2)} \delta(\omega + \omega') d\omega d\omega' \\ &= \frac{1}{2\pi} \int \frac{1}{m(\omega^2 + \Omega^2)} d\omega \\ &= \frac{1}{2m\Omega} \end{aligned} \quad (10)$$

Plugging (10) back into (3) with Wick's theorem in mind yields:

$$Z_\beta = Z_0 \left(1 - \int_{-\infty}^{\infty} \frac{3g}{4m^2\Omega^2} d\tau' + \dots \right) \quad (11)$$

Here we recall that the limits of the time integral were taken to ∞ only for the purposes of the Fourier transform, and we now roll it back to the finite time interval over β to find:

$$\begin{aligned} Z_\beta &= Z_0 \left(1 - \frac{3g\beta}{4m^2\Omega^2} + \dots \right) \implies \ln Z_\beta \approx \ln Z_0 - \frac{3g\beta}{4m^2\Omega^2} \\ &\implies E_0 = - \lim_{\beta \rightarrow \infty} \frac{1}{\beta} \ln Z_\beta \approx \boxed{E_0^{(0)} + \frac{3g}{4m^2\Omega^2}} \end{aligned} \quad (12)$$

We now turn our attention to finding $E_0^{(0)}$, which we obviously expect to be $\Omega/2$. Normally this part is done first, but I've put it here to hopefully avoid some confusion with the way the path integral is being evaluated.

To begin with, we recall (1), with $V(q) = m\omega^2 q^2/2$. Our goal will be to find an expression we can work with inside of the integral over q_a . We won't be evaluating the exact path integral but this can be done to yield a precise expression, for instance see Swanson (4.23).

One way to make things simpler is to expand the path $q(\tau)$ into the classical path $q_{cl}(\tau)$, with an added deviation $x(\tau)$. The details of this are covered in Swanson Section 3.3, and we note in particular the result (3.43), reproduced here (note that Z is the path integral, not a partition function here):

$$Z(q_a, t_a, q_b, t_b) = F(t_a, t_b) e^{iS_c} \quad (13)$$

where S_c is the action evaluated for the classical path, and $F(t_a, t_b)$ is a quantity which does not depend on q_a, q_b . The utility is immediately obvious: this $F(t_a, t_b)$ can be pulled out of the outer integral over q_a in (1), and evaluated separately. And, in fact, after a Wick rotation, using the classical path for the harmonic oscillator, the exponential term in (13) is simply a Gaussian in q_a , and evaluating this portion independently yields a term of order $O(1)$.¹

The primary object of our interest is the energy, which will take a logarithm of Z_0 and then divide it by an infinite value, and so multiplications by terms of order $O(1)$ are irrelevant, and we will simply ignore this factor. Thus, we are now interested in the object $F(t_a, t_b) \sim Z_0$, which for potentials with at most quadratic order is precisely given by (see Swanson (3.42) and the subsequent discussion):

$$F(t_a, t_b) = \int_0^0 \overline{\mathcal{D}}x \exp \left[\frac{i}{\hbar} \int_{t_a}^{t_b} dt \mathcal{L}(\dot{x}, x) \right] \quad (14)$$

After performing a Wick rotation, this is precisely just (1), but with the outer integral removed and setting $q_a = 0$. If we do the same Fourier transforms, we find that, for our purposes, we can get away with taking (8), removing the outer integral, and setting $q_a = 0$, and this is exactly what we'll do:

$$Z_0 \sim \int_0^0 \overline{\mathcal{D}}q \exp \left\{ -\frac{m}{2\pi} \int_0^\infty (\omega^2 + \Omega^2) q(\omega) q^*(\omega) d\omega \right\} \quad (15)$$

Now we discretize the action integral to a certain number of points, N . Again, if the quadratic coefficient matrix is denoted A , the result is $\pi^N / \det A$, so:

$$\ln Z_0 \approx C - \ln \det A = C - \text{Tr} \ln A \quad (16)$$

where C includes the factors of π and the outer integral over q_a . Now, A is diagonal, so the logarithm of A is just the logarithm of its elements, and the trace is simply the sum along the diagonal. Our particular A is once again given by (9), and after taking a logarithm we can write the total expression as:

$$\ln Z_0 = C - \beta \int_0^\infty \frac{d\omega}{2\pi} \ln \left(\frac{m}{2\pi} (\omega^2 + \Omega^2) \right) \quad (17)$$

¹Details about the classical path and using it in this way can be found in Swanson Eqs. (3.52-3.54), then see (4.23). To argue it more precisely the infinite β limit must be taken, otherwise the integral actually does contribute a significant order of β and cannot simply be ignored.

where we have taken the trace by using the usual definition of Matsubara frequencies $\omega_n = 2\pi n/\beta$.² We now take a derivative with respect to Ω^2 on both sides to find:

$$\frac{d}{d\Omega^2} \ln Z_0 = -\beta \int_0^\infty \frac{d\omega}{2\pi} \frac{1}{(\omega^2 + \Omega^2)} = -\frac{\beta}{2\pi} \cdot \frac{\pi}{2\Omega} = -\frac{\beta}{4\Omega} \quad (18)$$

Integrating w.r.t. Ω^2 now (noting that $\int 1/\sqrt{x} dx = 2\sqrt{x} + C$), we find:³

$$\ln Z_0 = C - \frac{\beta\Omega}{2} \implies E_0^{(0)} = -\lim_{\beta \rightarrow \infty} \frac{1}{\beta} \ln Z_0 = \frac{\Omega}{2} \quad (19)$$

as expected. And so we see:

$$E_0 = \frac{\Omega}{2} + \frac{3g}{4m^2\Omega^2} + O(g^2) \quad (20)$$

as we know from regular perturbation theory.

The way we've solved this problem naturally lends itself to the VPT approach as well.

2 VPT

We can repeat the argument the same way while putting in the VPT parameter much the same way as before, call it Ω' . Then the new unperturbed frequency is $\sqrt{\Omega^2 + \Omega'^2}$ and the perturbation we pull out in the calculation leading to (3) includes the term included due to VPT, i.e.:

$$Z_\beta = Z_0 \left(1 - \int_{-\infty}^{\infty} d\tau' \left[g \langle q(\tau')^4 \rangle - \frac{1}{2} m \Omega'^2 \langle q(\tau')^2 \rangle \right] + \dots \right) \quad (21)$$

Then we again just use (10) but with the adjusted value for Ω to find, similar to (11):

$$Z_\beta = Z_0 \left(1 - \int_{-\infty}^{\infty} \frac{3g}{4m^2 (\Omega^2 + \Omega'^2)} - \frac{\Omega'^2}{4\sqrt{\Omega^2 + \Omega'^2}} d\tau' + \dots \right) \quad (22)$$

and finally find an expression similar to (17):

$$E_0 \approx \frac{\sqrt{\Omega^2 + \Omega'^2}}{2} + \frac{3g}{4m^2 (\Omega^2 + \Omega'^2)} - \frac{\Omega'^2}{4\sqrt{\Omega^2 + \Omega'^2}} \quad (23)$$

Here we minimize the sensitivity w.r.t. Ω' , but this ends up being a cubic equation that we're better off just figuring out numerically. So let's set $\Omega = m = 1, g = 0.2$ and then plot the derivative, which yields Figure 1.

²That is, since the trace is the sum over n , and since the summand changes slowly w.r.t. n in the $\beta \rightarrow \infty$ limit, it can be made an integral over n , and then we can use $dn = \beta d\omega/2\pi$ to find the particular form of the trace in (17).

³It is a legitimate question to ask the nature of C , since our assumption is that it vanishes when divided by $\beta \rightarrow \infty$. We can be more careful and define bounds from $0 \rightarrow \Omega^2$, and then note that in the zero temperature limit the free particle partition function is just 1.

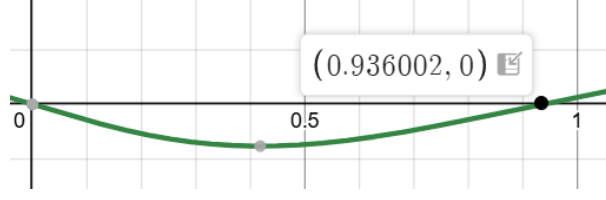


Figure 1: Derivative of (20) with Constants = 1, $g = 0.2$

Thus, $\boxed{\Omega' \approx 0.936}$ which we already know to be correct.

3 Diagrams

The goal here is to first write (20) as the result of Feynman diagrams, using the fact that (12) is just the linked cluster theorem. To do this, we need to do three things: 1) Figure out the general propagator between vertices at two different times, 2) Figure out what the vertices themselves represent, and 3) Figure out the symmetry factor.

Luckily all of these steps are straightforward. The first is simply $G(\tau, \tau') = \langle q(\tau)q(\tau') \rangle$ which we can compute. The second we can probably guess is just a factor of g , perhaps with a negative sign, and finally the symmetry factor is in fact what we've already found for the 0D field theory case, since here the fields are real.

Let's first figure out the propagator. This is given by:

$$\begin{aligned}
 \langle q(\tau)q(\tau') \rangle &= \left\langle \left\{ \frac{1}{2\pi} \int e^{-i\omega\tau} q(\omega) d\omega \right\} \left\{ \frac{1}{2\pi} \int e^{-i\omega'\tau'} q(\omega') d\omega' \right\} \right\rangle \\
 &= \left\langle \frac{1}{4\pi^2} \int \int e^{-i(\tau\omega + \tau'\omega')} q(\omega)q(\omega') d\omega d\omega' \right\rangle \\
 &= \frac{1}{4\pi^2} \int \int e^{-i(\tau\omega + \tau'\omega')} \langle q^*(-\omega)q(\omega') \rangle d\omega d\omega' \\
 &= \frac{1}{4\pi^2} \int \int e^{-i(\tau\omega + \tau'\omega')} \frac{2\pi}{m(\omega^2 + \Omega^2)} \delta(\omega + \omega') d\omega d\omega' \\
 &= \frac{1}{2\pi m} \int \frac{e^{-i\omega(\tau - \tau')}}{(\omega^2 + \Omega^2)} d\omega
 \end{aligned} \tag{24}$$

This is a simple integral to evaluate using contour integration, particularly invoking the residue theorem and Jordan's lemma. For $\tau > \tau'$, the contour is closed below, and for $\tau < \tau'$ the contour is closed above. In the former case the enclosed pole is $\omega = -i\Omega$ whereas in the latter case it is $\omega = i\Omega$. Jordan's lemma tells us that the integral over the semi-circular arcs vanish, and so the integral in (24) is simply equal to $2\pi i$ multiplied by the residue at the pole. It's straightforward to show at this point that, after evaluating the residues:

$$\langle q(\tau)q(\tau') \rangle = \frac{1}{2m\Omega} e^{-\Omega|\tau - \tau'|} \tag{25}$$

Just to briefly comment on the specifics, we first separate the denominator of the integrand into its poles and then evaluate them, and this evaluation gives a factor of $2i\Omega$

in the denominator and the exponential in (25). In calculating the residues there is a difference between the two poles by a minus sign, but in closing the contour below the loop is clockwise so the negative sign cancels, and the $2\pi i$ that comes with the residue theorem takes care of the i in the denominator and the 2π sitting outside the integral in (24).

At this point the diagrammatic approach practically falls out of this formulation. We interpret lines between vertices by (25), each vertex as a factor of g , calculate the symmetry factors as in the 0D case, and we are set.

The first-order term is trivial using (25) because the time difference is zero, so let's instead consider second-order. We recall from 0D field theory that there are only two topologically distinct diagrams at second-order, pictured below.



(ii) Second Order

You can think of time as going left-to-right, but in any case the point is that we use (25), and then integrate over both times as usual. The first diagram has a symmetry factor of 12, the second has a symmetry factor of 36. So:

$$\begin{aligned} -\beta E_0^{(2)} = & 36 \times \left\{ \frac{g^2}{(2m\Omega)^4} \int \int e^{-2\Omega|\tau-\tau'|} d\tau d\tau' \right\} \\ & + 12 \times \left\{ \frac{g^2}{(2m\Omega)^4} \int \int e^{-4\Omega|\tau-\tau'|} d\tau d\tau' \right\} \end{aligned} \quad (26)$$

where the time integrals are over the entire real line. This is not too difficult to evaluate and yields the familiar result:

$$E_0^{(2)} = -\frac{21g^2}{8m^4\Omega^5} \quad (27)$$

We now venture to write VPT using diagrams, and this is actually not very difficult to do. Indeed, the idea is once again to separate the Hamiltonian like so:

$$\hat{H} = \underbrace{\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 + \frac{1}{2}m\Omega^2\hat{x}^2}_{\hat{H}_0} + \underbrace{g\hat{x}^4 - \frac{1}{2}m\Omega^2\hat{x}^2}_{\hat{H}_1} \quad (28)$$

Note that I'm going to be switching notational convention here: originally ω was saved for the Fourier frequency, Ω was the oscillator frequency, and Ω' was the VPT parameter. But now, we go back to treating ω as the oscillator frequency, and Ω as the VPT parameter.

Using Wick's theorem we see that the perturbation in (28) will consist of four-legged vertices and two-legged vertices. We might preliminarily expect the two-legged vertex to simply encode a factor of $-m\Omega^2/2$, and as we'll see this gives us exactly (23). The present goal is to figure out a way to write down VPT diagrams systematically, and trying to figure out if the minimization procedure has a graphical analogue.

Let's first start by writing down the first-order VPT diagrams and confirming that they give you (23). This is not very hard to do, indeed the only topologically distinct diagrams are:



(i) First Order

where we note that the diagram on the left has a 4-legged vertex (i.e. the quartic vertex), as opposed to the diagram on the right which has a 2-legged vertex (i.e. the VPT vertex). We already know that the diagram on the left yields the usual first-order correction to the quartic anharmonic oscillator via perturbation theory, except that since we are dealing with $\hat{H}_0$ as defined in (28), the propagators have a shifted frequency $\tilde{\omega} \equiv \sqrt{\omega^2 + \Omega^2}$. This already explains the first two terms of (23) (again, note the change of notation).

Now, the diagram on the right has a vertex we interpret as $-m\Omega^2/2$ and a propagator we interpret as $1/2m\tilde{\omega}$. Multiplying the two, we find this diagram evaluates to $-\Omega^2/4\tilde{\omega}$, which is precisely the last term in (23) (of course, recall that we should really first integrate over time and then divide by β , but since there is no time dependence the integral and the factor of β just cancel).

This is not too interesting so far, so let's go fishing for a diagrammatic version of the PMS condition, which in this case is as simple as asking where the derivative of the energy w.r.t. the VPT parameter vanishes. At the moment I only have this worked out for 1o, which is what we'll consider here.

The PMS condition can be stated as follows:

$$\frac{dE_0}{d\Omega} = 0 \quad (29)$$

And diagrammatically we can write the energy (at 1o) as:

$$E_0 = \bigcirc + \bigcirc \bullet \bigcirc + \bullet \bigcirc \quad (30)$$

where I'm using the tiny circle to represent the zeroth-order energy. Again, (30) is with the understanding that the time integration cancels out with the factor of β . We now ask what the derivative w.r.t. the VPT parameter does to (30).

We see that taking the derivative on the free propagator yields a rather suggestive form: $1/2m\tilde{\omega} \rightarrow (1/2m\tilde{\omega})^2(-m\Omega^2/2) \cdot (4/\Omega\tilde{\omega})$. So essentially this is the same as taking every line, and putting a VPT vertex in the middle of it (up to a factor of $4/\Omega\tilde{\omega}$).

Taking the derivative on the "pure bubble" (i.e. first diagram in (30), i.e. $\tilde{\omega}/2$) yields, as you can show, $-(\tilde{\omega}/2)(1/2m\tilde{\omega})(-m\Omega^2/2)(4/\Omega\tilde{\omega})$.⁴ So, this is a just the pure bubble multiplied by the first order VPT diagram (up to a negative sign and again, a factor of $4/\Omega\tilde{\omega}$).

The quartic vertex is unaffected by the derivative. However, the VPT vertex is affected, and when differentiated yields $-m\Omega^2/2 \rightarrow (-m\Omega^2/2)(\tilde{\omega}/2)(4/\Omega\tilde{\omega})$, which is a pure bubble multiplied by the VPT vertex, again up to the same factor.

⁴I am writing these results in a weird way particularly to pull out a ubiquitous factor of $4/\Omega\tilde{\omega}$, and putting everything in the language of the basic diagram components we know.

This is a massive pain to write on using tikz so I'm simply going to post a screenshot of my handwritten work to summarize what we're getting at below.

$$\begin{array}{c}
 \text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} \\
 \downarrow \text{remove implicit symmetry factors.} \quad \partial/\partial \Omega \cdot \frac{\Omega \tilde{\omega}}{4} \\
 -\cancel{\text{Diagram 1}} + \text{Diagram 4} + 6 \text{Diagram 5} \\
 \quad \quad \quad + \cancel{\text{Diagram 6}} = 0 \\
 \Rightarrow \text{Diagram 7} + 6 \text{Diagram 8} = 0 \\
 \text{or, } -\frac{1}{2} m \Omega^2 = -\frac{6g}{2m\tilde{\omega}} \\
 \Rightarrow \Omega^2 = \frac{6g}{m^2 \tilde{\omega}}
 \end{array}$$

Figure 2: VPT PMS Condition as Diagrams

As noted, we are removing the global factor (since we're setting everything to zero anyway), but importantly we're also removing the implicit symmetry factors. There is a systematic way to deal with the symmetry factors but I won't consider that here. Note that we are simply plugging in vertices into lines (with vertices on the same vertical position to signify the fact that they are not at different times), and using the product rule in going from the VPT diagram to the second and fourth diagrams in the derivative, and in going from the quartic 1o diagram to the third diagram (and its prefactor). The first and fourth diagrams cancel, and similar factors in the second and third diagrams drop out to find the final equation that defines the VPT parameter (field).

This has a very particular form: indeed, the VPT parameter is essentially just Hartree-Fock (i.e. sum of all bubble diagrams, but in this case the logarithm makes it just the one bubble). Thus, the propagator itself encapsulates Hartree-Fock (as the VPT field is in the propagator).